1. **Introduction to Cosmology**: This module serves as the foundational stepping stone, familiarizing students with astronomical scales, structures of the universe and observational datasets. It is divided into eight interactive panels:

   - **Panel 1 - Structures of the universe:** Features a gamified drag-and-drop "card board" where students match celestial objects (e.g., comets, exoplanets, galaxies) to their correct physical dimensions. The interactive game includes a real-time scoring system (+1 point for a correct match, -0.25 penalty for a wrong pair), a timer and flip-card mechanics. Additionally, the panel includes a unit conversion information box to familiarize students with astronomical distances (e.g.,kilometers, astronomical units (AU), light-years (ly) and parsecs (pc)).
   - **Panel 2 and 3 - Discoveries and universe timelines:** Two interactive historical panels that trace the "history of astronomical discoveries" that covers eras from Prehistory to the modern Space Age, and the history of the universe (from the Big Bang to the present day) that distinguishes between theoretical epochs (e.g., Planck Era, Inflation) and observational epochs. Both panels feature an interactive quiz-game where students must identify the correct event epoch based on scientific descriptions. The quiz incorporates a timer and dynamic scoring system. When students answer correctly, they unlock an "explore topic" feature, providing specific windows with images and scientific references.
   - **Panel 4 - Observational instruments:** Provides an overview of the major ground-based and space-based telescopes, as well as observational methods like spectroscopy, photometry and interferometry. It explains the role of detectors such as CCDs (Charge-Coupled Devices). The panel shows an interactive simulation where students adjust sliders to modulate telescope diameter and exposure time, connected to light-gathering power and signal-to-noise ratio. Optimizing these settings, there is an impact on image resolution (number of pixels), resulting in the detection of the correct galaxy image.
   - **Panel 5 - Milky Way:** Explores the structure of our home galaxy. This panel focuses on galactic geography, featuring an interactive activity where students must investigate the galaxy's layout and find the correct position of the Solar System within the Milky Way.
   - **Panel 6 - Galaxies:** Students visualize the spatial distribution and evolution of galaxies. This panel integrates pre-computed animated visualizations from the Sloan Digital Sky Survey (SDSS) dataset [?] and generated via background Python scripts using `Imageio` [?] and `Astropy` [?]. The animations dynamically display, ordered by the redshift, the spatial distribution of galaxies mapping Right Ascension (RA) versus Declination (Dec). Furthermore, it plots their morphological evolution by analyzing the concentration index ($R90/R50$) versus the color index ($u-r$). This clearly demonstrates how galaxy populations, categorized into Elliptical (red, high concentration), Lenticular, Spiral and Irregular (blue, low concentration), change over cosmic time [?]. Users can also interact with `Plotly` [?] charts to click on individual galaxies and inspect their morphological data. Moreover, the panel shows a "Memory-Game" where users need to find the right couples of galaxies based on their morphological classification.
   - **Panel 7 - Stars:** Explores stellar evolution using data from the Gaia mission and MIST isochrones [?, ?]. This panel features a game to track a star evolution starting from its mass. Moreover, it shows an animated pre-built Hertzsprung-Russell (HR) diagram tracking absolute magnitude against color ($BP-RP$), dynamically calculated using Gaia's parallax data and apparent magnitude. The visualization demonstrates stars transition (ordered by their age) through distinct evolutionary phases: Main Sequence (MS), Subgiant Branch (SGB), Red Giant Branch (RGB), Helium Burning (HeB), Asymptotic Giant Branch (AGB) and White Dwarfs (WD).
   - **Panel 8 - Fundamental particles:** Explores the subatomic world and the Standard Model of particle physics, connecting the micro-cosmos to the macro-cosmos. The panel shows the particle classification challenge, a gamified "drag-and-drop" game where students must correctly sort fundamental particles (e.g., e^-, u, d, γ, p, n) into their proper categories

(Fermions, Leptons, Quarks, Bosons, Hadrons, Mesons, Baryons) under a time limit, with a dynamic scoring system. To assist them, students can explore interactive network graphs detailing particle properties and interactions, as well as an interactive plot mapping particles ordered by their masses. Finally, the panel includes mass-energy conversion box allowing students to learn how to convert rest mass in kilograms to energy in electronvolts using Einstein's mass-energy equivalence ($E = mc^2$), which is useful for understanding the cosmological processes of the early Universe (e.g., Big Bang Nucleosynthesis, Recombination, Reionization).

2. **Dark Matter** Investigates the missing mass problem across different cosmic scales.

   - **Panel 1 - Kepler's Laws :** A preparatory panel designed to reinforce the classical relationship between mass, distance and orbital velocity before transitioning to galactic scales.
     - *Planets dataset:* Students interact with NASA/JPL planetary ephemerides data [**?**, **?**, **?**, **?**]. The panel includes a built-in file management system allowing students to download the raw dataset, perform their velocity calculations ($v = \sqrt{GM/r}$) in an external spreadsheet environment and upload their completed exercises directly to the application's cloud storage.
     - *Solar System model:* The application plots the calculated velocities against the semi-major axes of the planets. It visually confirms the Keplerian decline in velocity as distance increases, establishing the baseline expectation for systems dominated by baryonic mass. It also features a supplementary plot verifying Kepler's third law ($P^2 \propto a^3$).
   - **Panel 2 - Galaxy rotation curves:** This is the core inquiry panel where students recreate Vera Rubin's discovery testing different galaxies trough SPARC dataset [**?**].
     - *Velocity discrepancy:* Students examine the observed spectroscopic rotational velocities (blue dots) against the radial distance from the galactic center. They compare this to the theoretical Keplerian curve (red curve) calculated from the quadratic sum of the gas, disk and bulk velocity contributions [**?**]. The plot shows an anomaly: the theoretical curve decays, but the observed data remains inexplicably "flat."
     - *Interactive exercise:* To fix the model, students manually inject a Navarro-Frenk-White (NFW) density profile [**?**, **?**, **?**, **?**, **?**], by manipulating a slider that adjust the dark matter mass. They dynamically move a simulated velocity curve (green curve) to match the observational data.
     - *Statistical validation:* Rather than guessing visually, the application provides a mathematical tool to quantify the discrepancy between the simulated curve and the observational data. Students calculate the χ^2 goodness-of-fit choosing different dark matter mass estimates. The app plots these attempts point-by-point, constructing a minimization parabola whose vertex reveals the exact mass of the dark matter and the minimum distance between the two curves. The choice to employ parabolic minimization is also connected to modern applications of the same concepts, for instance the logic—loss function optimization, that is the basis of artificial intelligence and machine learning algorithms. Additionally, a top-down visual representation of a galaxy morphology, connected to the slider, color-codes the mass distribution: the central black hole, the baryonic bulge/disk and the extended dark matter halo.
     - *Data sonification:* The panel features also a multisensory tool that maps velocity arrays to audio frequencies (e.g.,high values connected to high-pitched sound), allowing students to "hear" the alignment of the simulated curve with the real observations.
   - **Panel 3 - Galaxy mass and dark matter:** An analytical reinforcement panel that bridges the visual rotation curves with their underlying mathematics.
     - *Formula-building exercise:* Students, selecting a galaxy dataset, actively construct the physical models by filling in missing variables in interactive equation fields. They must

correctly build the formulas for total mass, baryonic mass components and theoretical Keplerian velocity. The application evaluates their inputs in real-time and if correct, generates comparative mass distribution and velocity plots.

- **Panel 4 - Cluster velocity distribution:** This panel expands the dark matter hypothesis to the largest gravitationally bound structures in the universe: clusters of galaxies.
  - *Dynamic visualization:* Utilizing spectroscopic datasets from massive galaxy clusters (e.g., Coma and Abell clusters) [**?**, **?**], students explore the kinematics of galaxies moving within the intra-cluster medium. To enhance student engagement and mimic the authentic process of histograms building and astronomical data acquisition, the application employs a real-time animated plotting mechanism. Using an asynchronous internal timer, the velocity data points are injected sequentially, interactively building both a phase-space scatter plot and a velocity distribution histogram point-by-point. Inside the code, once the histogram is fully populated, a Gaussian fit is mathematically applied to show the distribution, calculating the mean recession velocity (which determines the cluster's overall cosmological distance via Hubble's Law) and the line-of-sight velocity dispersion.
  - *Dark matter exercise:* The plots show observational data in blue, the baryonic velocity model in red (composed only of stars and gas contributions) and the dark matter simulation in green built from the adapted NFW density profile [**?**, **?**, **?**, **?**]. Students can move the dark matter slider in order to add the correct amount of additional mass to match the observations. In addition, the exercise for χ^2 minimization is available with the same logic as for galaxies, but considering the distance between the mean velocity observed and simulated.
- **Panel 5: Cluster mass and dark matter:** Focuses on utilizing the Virial theorem to estimate the total dynamic mass of galaxy clusters.
  - *Virial theorem application:* Through a guided pseudo-code exercise, students must input missing variables to properly reconstruct the Virial theorem equations from the kinetic energy and the gravitational potential of a simplified Newtonian system characterized by circular motion (gravitational and centripetal forces). Using the velocity dispersion (standard deviation) calculated in Panel 4 and the characteristic radius, the application computationally evaluates the total mass of the cluster.
  - *Density and mass plots:* The panel generates comparative plots of the radial density profile and total enclosed mass. These plots contrast the luminous components (galaxies stars mass from magnitude and luminosity [**?**, **?**] and hot X-ray gas mass using a scaling law [**?**]) against the vastly larger total dynamic mass, providing definitive proof of dark matter.